

Model Selection Report

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Project Name: Diabetic Retinopathy Classification Web Application

1. Objective

The objective of model selection is to identify the most suitable deep learning architecture for Diabetic Retinopathy classification.

2. Models Considered

The following approaches were analyzed:

- Basic CNN Model
 - Transfer Learning (MobileNet / VGG16)
 - Custom Sequential CNN
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3. Evaluation Criteria

Models were compared based on:

- Accuracy
 - Training Time
 - Computational Efficiency
 - Overfitting Behavior
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4. Selected Model

A Convolutional Neural Network (CNN) model was selected because:

- Good accuracy performance
 - Moderate computational requirements
 - Suitable for deployment in Flask web app
 - Easy integration with TensorFlow
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5. Conclusion

The selected CNN model provides a balanced trade-off between accuracy and computational efficiency, making it suitable for real-time Diabetic Retinopathy detection.

